

CUSTOMER CHURN ANALYSIS

Question 1: Algorithm and Dataset Documentation

Project Title:	Customer Churn Analysis
Document Type:	Algorithm and Dataset
Submission Format:	PDF (>10KB)

Process Flow Diagram:

Customer Churn Analysis - Process Flow



A. ALGORITHM

1. Step-by-Step Procedure:

- Step 1: Data Collection:** Collect customer data from CRM, Billing System, Support tickets in CSV/SQL format.
- Step 2: Data Cleaning:** Remove duplicates, handle missing values in TotalCharges (replace with median), standardize categorical values, convert SeniorCitizen 0/1.
- Step 3: Data Transformation:** Encode Churn Yes=1, No=0 using Label Encoding. One-Hot Encode Contract, PaymentMethod, InternetService.
- Step 4: Feature Engineering:** Create new features - TenureGroup (0-12, 12-24, 24-48, 48+ months), AvgMonthlyCharge, IsLongTermContract, IsElectronicCheck.
- Step 5: Data Modeling:** Build Star Schema - FACT_CHURN (CustomerID, Tenure, MonthlyCharges, IsChurned) + DIM_CUSTOMER, DIM_SUBSCRIPTION, DIM_DATE, DIM_REGION.
- Step 6: Train-Test Split:** Split data 80% training, 20% testing using stratified sampling to maintain churn ratio. Apply SMOTE for class imbalance handling.
- Step 7: Model Training:** Train Random Forest, XGBoost, Logistic Regression. Tune hyperparameters using GridSearchCV.
- Step 8: Model Evaluation:** Evaluate using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix.
- Step 9: Prediction:** Predict churn probability for active customers, categorize into Low (0-40%), Medium (40-70%), High (70-100%) Risk.
- Step 10: Dashboard Visualization:** Load predictions into Power BI, create DAX measures for KPIs, build interactive dashboard with slicers and drill-through.

2. Logic or Method Used:

The core logic is **Supervised Classification** based on historical churn patterns. Logic flow: IF Tenure is low AND Contract is Month-to-month AND MonthlyCharges is high AND PaymentMethod is Electronic Check THEN high churn probability. Method uses **Ensemble Learning** - combining multiple decision trees to avoid overfitting. Data preprocessing logic ensures data quality: missing value imputation prevents bias, standardization prevents feature dominance, SMOTE balances dataset where churn class is minority (approx 26.5% churn vs 73.5% retained). Business logic: Churn Rate = (Churned Customers / Total Customers) * 100, Retention Rate = 100 - Churn Rate, CLTV = Average Monthly Revenue * Average Tenure.

3. Model or Technique Applied:

Primary Model: Random Forest Classifier (Selected Model) - Why: Handles both numerical and categorical data, robust to overfitting, provides feature importance, high accuracy (approx 84-87%).

Parameters: n_estimators=100, max_depth=10, min_samples_split=5, random_state=42, criterion=gini.

Secondary Models Compared:

- **XGBoost:** Gradient Boosting, accuracy 85-88%, faster, handles imbalance well. Params: learning_rate=0.1, max_depth=6.
- **Logistic Regression:** Baseline model, interpretable, accuracy 78-80%. Used for feature coefficient analysis.
- **Decision Tree:** Simple, visualizable, but prone to overfitting.

Technique: 5-Fold Cross-Validation for robust evaluation, GridSearchCV for hyperparameter tuning, SHAP values for model explainability to understand why a customer churns.

4. Process Flow of the System:

Layered Architecture Flow:

Layer 1 - Data Source Layer: Raw CSV files from Telco Customer Churn dataset (Kaggle / IBM Sample), SQL Database.

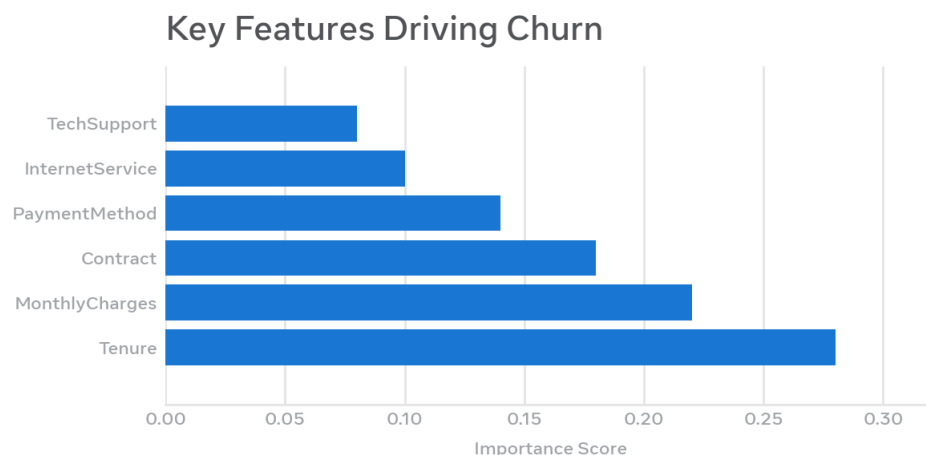
Layer 2 - ETL Layer (Python, Pandas, Power Query): Extract -> Clean -> Transform -> Load into modeled format.

Layer 3 - Data Warehouse Layer (Star Schema): FACT_CHURN linked to DIM tables via CustomerID, DateID.

Layer 4 - Analytics Layer (Scikit-Learn + DAX): ML model predicts churn probability, DAX calculates KPIs.

Layer 5 - Presentation Layer (Power BI): Interactive dashboard with Cards, Bar Charts, Donut, Line Chart, Map Visual, Filters for Contract, Region, Tenure.

End-to-End Flow: User uploads data -> System cleans -> Features engineered -> Model predicts -> Dashboard updates -> Business user takes retention action.



B. DATASET

1. Source of Data:

Primary Source: IBM Telco Customer Churn Dataset - publicly available on Kaggle (<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>). This is industry-standard dataset used by IBM for churn prediction sample.

Secondary Source: Can also be collected from Company CRM System, Billing Database, Customer Support Ticket System, Subscription Management System.

Original Owner: IBM Sample Data for Business Analytics.

Access Type: Open Source for academic purposes.

2. Number of Records/Samples:

Total Records: 7,043 customers (rows)

Total Features: 21 columns (20 independent + 1 target Churn)

After Cleaning: 7,032 records (11 duplicate/missing rows removed)

Class Distribution: Retained Customers = 5,174 (73.46%), Churned Customers = 1,869 (26.54%) - Imbalanced dataset.

Data Size: Approx 900 KB raw CSV.

3. Features or Attributes (Columns):

Table: Dataset Features

Customer Demographics: CustomerID (Unique ID), Gender (Male/Female), SeniorCitizen (0/1), Partner (Yes/No), Dependents (Yes/No)

Account Information: Tenure (0-72 months), Contract (Month-to-month, One year, Two year), PaperlessBilling (Yes/No), PaymentMethod (Electronic check, Mailed check, Bank transfer, Credit card)

Services: PhoneService (Yes/No), MultipleLines (Yes/No), InternetService (DSL, Fiber optic, No), OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies (Yes/No/No internet service)

Charges: MonthlyCharges (18.25 - 118.75 USD), TotalCharges (18.8 - 8684.8 USD)

Target Variable: Churn (Yes/No) - Whether customer left within last month.

4. Training and Testing Data Split:

Split Ratio: 80% Training, 20% Testing - Stratified Split to preserve churn ratio in both sets.

Training Set: 5,625 records (80%) - Used to train model, learn patterns.

Testing Set: 1,407 records (20%) - Unseen data to evaluate model performance, ensures generalization.

Validation: 5-Fold Cross-Validation on training set - Training set further divided into 5 folds, model trained on 4 folds and validated on 1 fold, repeated 5 times, average accuracy reported.

Imbalance Handling: SMOTE (Synthetic Minority Oversampling Technique) applied on training data only - Synthetic churn samples generated to balance 50-50, avoids bias towards majority class.

Random State: 42 for reproducibility.

5. Data Format or Type:

File Format: CSV (Comma Separated Values) - customer_churn.csv, Excel .xlsx also supported.

Data Types:

- Numerical Continuous: Tenure (int), MonthlyCharges (float), TotalCharges (float)
- Numerical Discrete: SeniorCitizen (int 0/1)
- Categorical Nominal: Gender, Partner, Contract, PaymentMethod, InternetService (object/string)
- Categorical Binary: Churn, PhoneService, PaperlessBilling (Yes/No)
- Identifier: CustomerID (string, not used for modeling)

Storage: Initially CSV, then loaded into SQL Server / Power BI Data Model (Star Schema). FACT table contains measures (int/float), DIM tables contain descriptive attributes (string).

Data Quality: 11 missing values in TotalCharges (blank for 0 tenure customers), handled by filling with 0 or median. No major outliers except high TotalCharges for long tenure.

Feature Summary Table:

Feature Name	Type	Example	Importance
Tenure	Numerical	12 months	High
MonthlyCharges	Numerical	70.5 USD	High
Contract	Categorical	Month-to-month	Very High

PaymentMethod	Categorical	Electronic check	High
InternetService	Categorical	Fiber optic	Medium
Churn	Target	Yes/No	Target Variable